

Caroline Piaulet-Ghorayeb

51 PEGASI B FELLOW (2026) & MARGARET BURBIDGE PRIZE POSTDOCTORAL FELLOW · UNIVERSITY OF CHICAGO

✉ carolinepiaulet@uchicago.edu | 🏠 cpiaulet.github.io | 📄 github.com/cpiaulet

General Information

- Nationality: French
- Languages: French (Native), English (Fluent), German (basics)
- Affiliations: University of Chicago Astronomy & Astrophysics Dept., Canadian Astronomical Society (CASCA)
- Research Expertise: Exoplanet atmosphere modeling and retrievals; Planet structure models; Atmosphere observations (from space & ground); Development of modeling, data reduction and analysis tools; RVs, TTVs.

Academic Appointments

51 Pegasi b Postdoctoral Fellowship - Heising-Simons Foundation

Chicago, IL, USA

UNIVERSITY OF CHICAGO

Sep 2026-

- Project Title: Revealing the Presence and Origins of Water on Small Exoplanets
- Duration: 3(+1) years

Margaret Burbidge Prize Postdoctoral Fellowship

Chicago, IL, USA

UNIVERSITY OF CHICAGO

Sep 2024-

- Project Title: Seeing the Unseeable: Opening new windows into exoplanet atmospheres and compositions
- Duration: 3 years

Dept. of Education, University of Helsinki

Helsinki, Finland

VISITING RESEARCHER (EDUCATION & NEUROSCIENCE)

Spring 2025

- Copernicus Project (PI: Kirsi Tirri)
- Application of machine learning algorithms to student brain responses and association with growth/fixed mindset (Puusepp, Piaulet-G., et al., in prep.)

Education

Université de Montréal

Montréal, Canada

PH.D. ASTROPHYSICS

Jan 2019 - Sep 2024

- Advisor: Prof. Björn Benneke
- GPA: 4.30/4.30; Thesis received top “Exceptional” mention.

Université de Montréal

Montréal, Canada

B.Sc. PHYSICS

Sept 2015 - Dec 2018

- Undergrad thesis advisor: Prof. Björn Benneke
- GPA: 4.30/4.30; 2016, 2017, 2018 Dean's Honor's List; Georges-Baril B.Sc. Excellence Prize

Student Supervision

SUPERVISED 1 GRAD STUDENT, 3 UNDERGRADUATE STUDENTS, 4 HIGH-SCHOOL STUDENTS, 1 PUBLICATION AND 2 IN-PREP *HS: HIGH SCHOOL STUDENT, UG: UNDERGRADUATE STUDENT; GS: GRADUATE STUDENT

- | | |
|--------|--|
| 2026 | Emily Mirkin (HS) , Open-source software development for sub-Neptune composition Inference. Received 1st Place Prize in the Somers Science Fair Competition for this project. |
| Summer | Adrien Donin De Rosiere (UG) , Machine learning for the mitigation of stellar contamination |
| 2026 | in exoplanet spectra. Received the College Funding Award for this project. |
| | Annabel Yarborough (UG) , Data reduction and atmosphere modeling for HST/UVIS eclipse |
| 2025- | dataset. Quad Research Grant and Travel Grant recipient for this project, Poster at Quad |
| | Symposium, Selected for a Contributed Talk at the ERES IX Conference (Ann Arbor, MI) |
| 2025- | Fernanda Correa Horta (GS) , Modeling of water world interiors and atmospheres. |
| 2023 | Elyse D'Aoust (UG) , Summer internship: modeling of giant planet reflected light |

- 2022-2023 **Danika Belzile, Mehramat Kaur, Kamrul Musfirat (HS)**, Modeling of transit-timing variations.
Danika Belzile, is now a summer intern at the Institute for research on exoplanets, and a Trottier Excellence Scholarship recipient.
Mehramat Kaur, is currently a “Climate75” Fellow. Kamrul Musfirat is the recipient of a L’Oréal Canada & Technovation scholarship, and a research assistant in the Fritz Lab at McGill University.
Kamrul Musfirat, is now the recipient of a L’Oréal Canada & Technovation scholarship, and a research assistant in the Fritz Lab at McGill University.
- 2026 **Team Mentor: Hack-4-Sages Astrobiology Hackathon**, Team Project: Distinguishing Biosignature Candidates from an Abiotic Baseline
- 2018 **Initiative XENOS (Solar Ecosystem contest)**, Scientific advisor and mentor in physics for the “Solar Ecosystem” contest (winning team!).

Teaching Experience

- 2025 **Invited Lecturer**, Theoretical Astrophysics Program (TAP), University of Arizona
 2025 **Invited Lecture**, ExoSLAM Workshop: Demystifying Atmospheric Retrievals
 2021 **Teaching Assistant**, PHY 6795P: Modern Analysis of Physical Data (Grad. level)
 2020 **Tutor**, PHY 1620: Physics of waves and vibrations (B.Sc. level)
 2019-2020 **Teaching Assistant**, PHY 2701: Astron. & Astrophys. (B.Sc. level)
 2019-2024 **Workshop Designer & Facilitator**, Projet SEUR / 1000 Sciences: crafting science and astronomy workshops for Cegeps and secondary schools
 2022-2023 **Conception and supervision of a mentored 8-week research project for gifted primary school students**, Initiation to programming for astrophysics. Offered to 2 cohorts.
 2022-2024 **Workshop: the scientific method applied to a research project**, Workshop presentation for high school students delivered in > 10 classrooms.
 2021-2022 **Conceive and deliver series of presentations on the scientific method**, Supported by iREx and acceSciences. Presented at the Vanier cegep.
 2020-2021 **Workshop animation and conception: “Science Sprints”**, 2-hr online presentations, high-school classrooms; over 800 students. Supported by the *Coeur des Sciences de l’UQAM*
 Jan 2020 **Physics Research Initiation Day**, Introduction to Python programming for cegep students.

Other Research Experience

Center for Space and Habitability, University of Bern

Bern, Switzerland

VISITING INTERN (EXOPLANETS, THEORY AND MODELING)

Summer 2019

- Supervisor: Prof. Kevin Heng
- Physically-motivated optical phase curve modeling and retrieval code (Morris, Heng, Jones, **Piaulet** et al. 2022)
- FRQNT International Internship Program Scholarship

Université de Montréal, iREx

Montréal, Canada

PHY 3030 (UNDERGRADUATE THESIS)

Fall 2018

- Supervisor: Prof. Benneke
- Analysis of WASP-107 b’s *Spitzer* eclipse and transits
- Presented at the 2019 **Extreme Solar Systems IV** Conference

Université de Montréal, iREx

Montréal, Canada

TROTTIER SCHOLAR (SUMMER INTERNSHIP)

Summer 2018

- Supervisor: Prof. Benneke
- Implementation of a new temperature structure retrieval method
- Resulting publication: Pelletier, Benneke, Darveau-Bernier, Boucher, Cook, **Piaulet** et al., 2021

Université de Montréal, CERN/ATLAS Group

Montréal, Canada

PARTICLE PHYSICS INTERN

Summer 2017

- Supervisors: Prof. J.-F. Arguin, Kazuya Mochizuki
- Implementation of machine learning algorithms to reduce charge misidentification in CERN/ATLAS data (**Piaulet et al.**, 2017; CERN-COMM)

Université de Montréal

Montréal, Canada

ASTROPHYSICS INTERN (MASSIVE STARS, OBSERVATIONS)

Summer 2016

- Supervisor: Prof. Nicole Saint-Louis
- Spectroscopic data reduction & analysis, stellar binary system WR137 (St-Louis, **Piaulet et al.**, 2021)

International Collaborations

ISSI Cool Worlds Proposal Team

LEAD FOR SUB-NEPTUNE ATMOSPHERES

2026-

- Interdisciplinary team of 12 researchers worldwide (planet detection and validation, mass measurements, demographics, interiors and atmospheres)
- Lead of aerosol-survey project on temperate sub-Neptunes.

Aether Collaboration

TEAM MEMBER, SUB-NEPTUNE INTERIOR-ATMOSPHERE MODELING EXPERT

2026-

- Carnegie-led collaboration on projects bridging laboratory experiments, geophysics, interior and atmosphere modeling and observations to understand sub-Neptune compositions.

Water Worlds JWST Proposal Team

LEAD ON GJ 9827 D DATASET, ATMOSPHERE MODELING LEAD ON GJ 3090 B DATASET

2022-

- Interdisciplinary team of 30+ researchers worldwide (planet formation, interiors, evolution, data analysis experts)
- Lead author of telescope proposal, team meeting lead (2023-2024).
- Data reduction, analysis, modeling, interpretation for GJ 9827 d (lead); atmosphere modeling for other datasets

Early Release Science (Exoplanet Transits)

NIRSPEC PRISM DATA REDUCTION + CO-LEAD OF “CHEMICALLY-CONSISTENT” SUBGROUP

2022-

- Interdisciplinary team of over 340 researchers worldwide
- Contributed main analysis of JWST/NIRSpec PRISM data
- Co-Lead of the “Chemically-Consistent” subgroup for atmosphere modeling and characterization

NEAT JWST Atmosphere Characterization team

LEAD ON TRAPPIST-1 D DATASET, ATMOSPHERE MODELING LEAD ON WASP-107 B DATASET

2022-2025

- Interdisciplinary team of over 40 researchers (PI: Prof. Lafrenière); Canadian GTO program
- Leading reduction, analysis, modeling, interpretation for TRAPPIST-1 d observations
- Atmospheric retrievals, stellar contamination studies, data reduction, interpretation on other datasets

NIRPS: Near Infra Red Planet Searcher

SUB-PROGRAM CHAMPION

2022-2023

- 3.6m Telescope at La Silla Observatory (ESO, Chili)
- Champion of the scientific proposal for atmospheric characterization of small planets in multiplanetary systems.

Selected PI and Lead Author Observing Proposals

12 SUCCESSFUL PROGRAMS ON SPITZER, CFHT/SPIROU, HST, JWST C1,2,3,4; INCL. 5 PI AND 3 LEAD AUTHOR

James Webb Space Telescope Cycle 4 Proposal (Program 9095), “Combining Emission and Transmission Spectroscopy to reveal Exo-Neptune Aerosols, Chemistry, and Formation”, 2025, 38 hours (**PI: Piaulet-Ghorayeb**).

James Webb Space Telescope DDT Proposal (Program DD 12492), “Confirming an 8th planet in the TRAPPIST-1 system”, 2026, 23 hours (PI: Agol; Piaulet-Ghorayeb Co-I).

James Webb Space Telescope Cycle 4 Proposal (Program 9101), “Unveiling the Nature of Super-Puffs: A Panchromatic Transmission Spectroscopy Survey”, 2025, 95 hours (**PI Piaulet-Ghorayeb**).

James Webb Space Telescope Cycle 4 Proposal (Program 8696), “Ammonia and methane on a temperate giant planet orbiting a late M dwarf”, 2025, 7 hours (PI: Zhang, Piaulet-Ghorayeb Co-I).

James Webb Space Telescope Cycle 3 Proposal (Program 6457), “Thermal emission of a cool, potentially volcanically active exo-Earth”, 2021 (PI: Benneke, Piaulet-Ghorayeb lead author).

James Webb Space Telescope Cycle 3 Proposal (Program 5967), “Exploring the desert: Thermal characterization of an exposed planetary core”, 2023 (PI: Roy, Piaulet-Ghorayeb Co-I).

James Webb Space Telescope Cycle 2 Proposal (Program 4098), “Exploring the existence and diversity of volatile-rich water worlds”, 2023, 82 hours (PI: Benneke, Piaulet lead author).

Hubble Space Telescope Cycle 30 Proposal (Program 17090), “Anchoring the Energy Budget of the Keystone Jwst ERS Ultra-Hot Target WASP-18 b”, 2022, 10 orbits (**PI: Piaulet**).

James Webb Space Telescope Cycle 1 Proposal (Program 2594), “The twin paradox: assessing planetary radius evolution with a CH₄ thermometer”, 2021 (PI: Spake, Piaulet Co-I).

SPIRou DDT Proposal (Program 20BD97), “Life on Venus? Mapping potentially biotic phosphine on our nearest neighbour”, 2020, 1 hour (**PI: Piaulet**).

SPIRou Proposal (Program 20AC039), “Spectroscopic Characterization of a Newborn Neptune-Sized Planet”, 2020 (**PI: Piaulet**).

Spitzer Space Telescope DDT Proposal (Program ID: 14256), “Thermal Emission Observations of the Keystone Exoplanet WASP-107b”, 2019, 12 hours (PI: Benneke, Piaulet lead author).

Selected Publications

7 FIRST-AUTHOR, 63 REF'D (+9 SUBM.), 3,700+ CITATIONS, H-INDEX=29, ORCID: 0000-0002-2875-917X

- **Piaulet-Ghorayeb, C.**, “Sub-Neptune Atmosphere Characterization in the JWST Era” (2026, Invited Review for **Nature Astronomy**, in revision)
- **Piaulet-Ghorayeb, C.**, et al., “A Window for Water-hydrogen Demixing on Warm Metal-rich Sub-Neptunes” (2026, ApJ, 1006, 59)
- **Piaulet-Ghorayeb, C.**, et al., “Strict Limits on Potential Secondary Atmospheres on the Temperate Rocky Exo-Earth TRAPPIST-1 d” (2025, ApJ, 989, 181)
- Luque, R., **Piaulet-Ghorayeb**, et al., “Insufficient evidence for DMS and DMDS in the atmosphere of K2-18 b: From a joint analysis of JWST NIRISS, NIRSpec, and MIRI observations” (2025, A&A, 700, A284)
- **Piaulet-Ghorayeb, C.**, et al., “JWST/NIRISS reveals the water-rich ‘steam world’ atmosphere of GJ 9827 d” (2024, ApJL, 974, L10)
- **Piaulet, C.**, et al. “Evidence for the volatile-rich composition of a 1.5-Earth-radius planet” (2023, **Nature Astronomy** 7, 206–222)
- **Piaulet, C.**, et al. “WASP-107b’s density is even lower: a case study for the physics of planetary gas envelope accretion and orbital migration” (2021, AJ, 161, 70)

Open-source Software

- **MELT**: Rocky planet energy balance (github.com/cpiaulet/melt); sole contributor.
- **SMINT**: Planet structure modeling (github.com/cpiaulet/smint); sole contributor.
- **EUREKA!**: JWST data reduction (github.com/kevin218/eureka); main contrib. (light curve extraction, fitting).
- **STCTM**: Stellar contamination modeling (github.com/cpiaulet/stctm); sole contributor.

Posters, Talks, Seminars and Lectures

40 TOTAL: 20 INVITED (2020:1; 2021:1; 2022:2 ; 2023:3; 2024:3, **2025:7, 2026:3**), 20 CONTRIB. (3 IN 2026)

Invited CANVAS Seminar (Canadian Astronomical Society-CASCA), 2026.

Contributed Plenary Talk, Exoplanets 6 Conference, Porto, 2026.

Contributed Talk, Canadian Astronomical Society-CASCA Annual Conference, Montreal, 2026.

Contributed Talk, Exoplanet Atmospheres Conference, Denver, CO, AASTCS 11, 2026.

Invited Astrophysics Seminar, University of Notre Dame, IN, 2026.

Invited Presentation, AGU/AbSciCon, Madison WI, 2026.

Invited Colloquium Speaker, Washington State University, 2025.

Ringberg Meeting, “Signal In the Noise”, Germany, 2025.

Invited Speaker, LIFE Space Mission Conference, Barcelona, Spain, “Rocky planet atmospheres: What we know & what’s to come”, 2025

Contributed talk, Atmospheric Escape and Replenishment in Planetary Systems Workshop, STScI, 2025.

Invited Seminar, Arizona State University, 2025.

Invited Colloquium, University of Arizona, 2025.

Invited Lecturer for the Theoretical Astrophysics Program (TAP) at the University of Arizona, 2025.

Contributed Talk at the EPSC-DPS25 Conference, Helsinki, “TRAPPIST-1 d: A Case Study for Atmospheric Loss at the Inner Edge of the Habitable Zone”, 2025.

Contributed Poster (ExoClimes VII conference), “Characterizing the volatile makeup of the 2-Earth-radii ‘steam world’ planet GJ 9827 d”, 2025.

Invited Lecture (ExoSLAM workshop), Demistifying Atmospheric Retrievals, 2025.

Invited Presentation, University of Helsinki (Copernicus team), “From stellar photons to students’ sweaty palms”, 2025.

Contributed Talk (Know Thy Star workshop), Caltech, 2025.

Invited Talk (MPIA Exocoffee), 2025.

Invited Seminar (ETH Zurich, exoplanet group), 2024.

Invited TAPS Seminar (Center for Space and Habitability, Bern), 2024.

Contributed Talk at The Great Link workshop, Max Plank Institute for Astronomy, Heidelberg, 2024.

Contributed Talk at the Extreme Solar Systems V Conference, 2024.

Invited Talk, CSH Seminar, Center for Space and Habitability, University of Bern, 2024.

Invited Seminar, UT Austin Astronomy & Astrophysics Department, 2023.

Invited ExoCoffee Seminar, University of Chicago Astronomy & Astrophysics Department, 2023.

Contributed Talk at First Year of JWST Science Conference, “Which small rocky planets have atmospheres?”, 2023.

Contributed Talk at Annual Conference of the CRAQ, 2023.

Invited Seminar for the Max Planck Institute for Astronomy-MPIA., 2023

Contributed Talk at TEPS Winter Conference, 2022

Contributed Talk at Annual Conference of the CRAQ, 2022. Awarded prize for one of the top student presentations.

Contributed Talk at the Exoplanets IV Conference, “Not all super-Earths are rocky planets”, 2022.

Invited Seminar: TEPS Graduate Fellows presentations, 2022

Contributed Talk at NIRPS Science Meeting, 2022

Invited Talk, JWST/NIRISS Science Team Meeting, “The JWST search for atmospheres around Trappist-1 d and f”, 2022.

“Three-Minute Thesis”-type Contributed Talk at TEPS Winter Conference, 2021

Contributed Talk at Annual Conference of CASCA, 2021.

Invited Talk at Annual Conference of CRAQ, 2021.

Invited Seminar: TEPS Graduate Fellows presentations, 2020

Contributed Talk at AAS 237 Conf., “Kepler-138d: a temperate Earth-mass planet that might be a water world”, 2020.

Contributed Talk at the Extreme Solar Systems IV Conference, “New insights into the keystone WASP-107 system”, 2019.

Contributed Talk at Annual Conference of CRAQ, 2019.

Contributed Talk at iREx interns final presentations, 2018.

Poster presentation at Canadian Conference for Undergraduate Women in Physics (CCUWiP), 2017.

Selected Distinctions: Prizes, Awards, Scholarships & Funding

7 INDEPENDENT FELLOWSHIPS INCL. HUBBLE & 51 PEG B; > \$ 700,000 IN GRANT FUNDING SINCE 2024

2026	2-yr Funding from International Space Science Institute (ISSI) , International Team – Project Title: “Cool Worlds: Demographics, Interiors, and Atmospheres of Long-Period Sub-Neptunes”	ISSI-Bern
2026	51 Pegasi b Fellowship (Accepted) , (3+1yr) \$265,717 + \$55,000 in research funds.	Heising-Simons
2026	NASA Hubble Fellowship -Declined , (3yr) \$262,500 + \$52,000 in research funds.	NASA
2026	\$ 10,000 , Suzuki Postdoctoral Fellowship	Y. Suzuki
2025	Courtois Institute Independent Postdoctoral Fellowship -Declined , (3yr, Montreal, QC, Canada) \$191,457 + \$90,000 in research funds.	Courtois Institute
2025	\$ 244,769 , Principal Investigator, STScI Project Grant; “Combining Emission and Transmission Spectroscopy to reveal Exo-Neptune Aerosols, Chemistry, and Formation”	STScI
2025	\$ 484,990 , Co-Principal Investigator, STScI Project Grant; “Unveiling the Nature of Super-Puffs: A Panchromatic Transmission Spectroscopy Survey”	STScI
2024	“Exceptional” Mention for PhD Thesis , Top mention for a doctoral thesis	UdeM
2024	Offered 4 Independent Postdoctoral Fellowships , Center for Space and Habitability (3yr; Bern, Switzerland; <i>declined</i>); STScI (4yr; Baltimore, MD, USA; <i>declined</i>); Trottier (2yr; Montreal, QC, Canada; <i>declined</i>); M. Burbidge Prize Fellowship, \$ 240,000 (3yr; Chicago, IL, USA; <i>accepted</i>)	CSH, Trottier, Brinson Foundation
2023	\$ 10,000 , First Prize of the “Impact Millénium” contest (for my NPO InitiaSciences); Entrepreneurship pitch competition across for-profit and non-profit organizations in Québec	Millénium Québecor
2023	\$ 500 , Engagement Award; Recognizes extraordinary community engagement	Physics Dept.
2023	Finalist, Forces Avenir contest as InitiaSciences (Category: Projects); Recognition of innovation, creativity and active leadership by young university students for a project or organization	Forces Avenir
2023	Finalist, Forces Avenir contest (Category: Personality, Graduate studies); Recognition of personal leadership, community impact and academic excellence	Forces Avenir
2022	\$ 50 , Second prize for Student Presentation; Annual Conference of the CRAQ	CRAQ
2022	\$ 2,000 , Laureate of ‘Osentreprendre’ Entrepreneurship contest in the Schools category; With my nonprofit, <i>InitiaSciences</i> . Recognizes innovative and creative organizations led by young students.	Osentreprendre
2021	\$ 72,000 , Excellence Scholarship for Foreign students; 3yr; Rank: 1 of 3 selected (<i>declined</i>)	NSERC
2021	\$ 84,000 , Bourse de Doctorat en Recherche; 4yr; Rank: 1 of 4 selected (<i>declined</i>)	FRQNT
2021	\$ 150,000 , Vanier scholarship; 3yr; Most competitive cross-disciplinary across Canada	NSERC
2021	Student Engagement Award , Recognition of engagement with youth, university, and the broader public in volunteering activities in parallel with academic excellence.	Fac. of Arts and Sciences
2020	\$ 5,000 , DIALOGUE; Science Communication Student Award, Co-created Educational Videos	FRQS
2019	\$ 2,000 , Georges-Baril Prize; Recognizes excellence of academic results throughout a BSc.	Fac. of Arts and Sciences
2019	\$ 100 , First prize for Student Presentation; Annual Conference of the CRAQ	CRAQ
2019	\$ 7,500 , FRQNT International Internship Program: Scholarship; Annual Conference of the CRAQ	FRQNT
2019	\$ 27,000 , TEPS scholarship; 3yr; Scholarship for excellent graduate students in exoplanet research.	TEPS/NSERC
2019	\$ 21,000 , FRQNT scholarship (B1X); 1yr; Direct bypass to PhD.	FRQNT
2019	\$ 20,000 , Excellence scholarship; 2yr; Direct bypass to PhD.	Physics Dept.
2018	\$ 6,000 , Trottier excellence scholarship; Summer internship at iREx.	iREx
2018	\$ 2,800 , Summer Internship Offer - DAAD RISE program; Competitive DAAD (German Academic Exchange Service) program (<i>Offer declined</i>)	DAAD
2016,17,18	Dean’s Honor List ,	Fac. of Arts and Sciences

Funding and Distinctions of Supervised Students

- 2026 **Contributed Talk at the ERES IX Conference, Ann Arbor, MI**, Annabel Yarborough
- 2026 **College Funding Award**, Adrien Donin de Rosiere
- 2026 **Poster at the UChicago Undergraduate Symposium**, Annabel Yarborough
- 2026 **Quad Undergraduate Research Scholarship**, Annabel Yarborough
- 2026 **Quad Undergraduate Research Conference Grant**, Annabel Yarborough

Additional professional training

EXTENSIVE TRAINING IN ATMOSPHERIC MODELING, TELESCOPE DATA ANALYSIS AND SCI-COMM.

- | | | |
|----------------|--|--------------------------|
| Mar 2022 | JWST Early Release Science (ERS-Transit) Data Challenge , Workshop to prepare for James Webb data analysis. | Baltimore, MD, USA |
| Jun 2021 | AstroComm 2021 Workshop , Training on astrophysics science communication | Online |
| May 2021 | ComSciCon-QC , Networking and expert training on science communication, participation only for selected applicants. | QC-Online |
| May & Oct 2021 | Calcul Québec Training Series , Compute Canada Spring School, and Calcul Québec JupyterLab training. | QC-Online |
| June 2019 | Exoclimes Simulation Platform (ESP) Summer School , Summer school, atmosphere modeling training. | Lenzerheide, Switzerland |

Service, Volunteering, Outreach

HIGHLIGHT: FOUNDED AND LED A HIGH-IMPACT NPO, 40+ TALKS TO PRIMARY, HIGH-SCHOOL CLASSROOMS AND AMATEUR ASTRONOMERS, 9 PRESS RELEASES, 30+ INTERVIEWS FOR TV, RADIO, WRITTEN PRESS

ASTRONOMY/PROFESSIONAL COMMUNITY SERVICE

SERVED ON TELESCOPE REVIEW PANELS, REFEREE FOR HIGH-IMPACT JOURNALS, CONFERENCE ORGANIZATION, DIVERSITY/CLIMATE COMMITTEES

- | | |
|----------------|--|
| Mar 2026 | Session Chair, Exoplanet Atmospheres 2026 Conference , Denver, CO |
| 2025-2026 | HST Exoplanets Panel & External Reviewer , |
| 2025 | Volunteer for mm Universe 2025 Conference , Local setup for conference, U Chicago |
| 2024- | Referee , AAS Journals, Astronomy & Astrophysics, Springer Nature Journals, MNRAS |
| Feb 2023 | EMAC Conference , Session chair and member of the organizing committee |
| Oct 2022 | TEPS Winter Conference , Session chair |
| 2026- | Canadian Astronomical Society (CASCA) Graduate Student Mentor , Provide academic career advice through monthly 1-1 meetings, feedback on application materials. |
| Aug – Dec 2020 | Organizing committee for the Canadian Space Summit 2020 , Organization and promotion of the “180 Seconds to a Degree” contest, contacting potential invited speakers, event planning; transition to online format (COVID-19). |
| Fall 2025- | UChicago Astronomy & Astrophysics Dept. Climate Committee , Coordinate climate survey preparation, dissemination and follow-up action plan. |
| 2022 - 2024 | iReX Diversity Committee , Planning of the institute’s activities to improve diversity, equity and inclusion in outreach activities and recruitment. |
| 2020–2024 | Physics Diversity Committee at UdeM - Shared leadership model , Lead and encourage conversations, organize invited talks from experts and action planning to promote EDI in recruitment, support mentoring of at-risk students. |

- 2020 – 2021 **Organizing committee member, Symposium-SAPHARI 2021**, Symposium for research and industry careers in Physics; event planning, sponsor contact, session chairing
- January 2020 **Kiosk during UdeM's Open House day**, Information about BSc, MSc and PhD programs for prospective Physics students.
- Dec 2021 **Facilitator for the ceremony "Celebrating Arts and Sciences" Université de Montréal, Canada**, Animation of the annual event for scholars of the Faculty of Arts and Sciences.

SUPPORTING YOUTH, WOMEN, GIRLS, AND UNDERSERVED COMMUNITIES

FOUNDED AND LED AN AWARD-WINNING NONPROFIT OPENING RESEARCH TO HIGH-SCHOOLERS; A DECADE OF KIOSKS, WORKSHOPS, PANELS AND MENTORING FOR GIRLS, YOUTH AND UNDERSERVED COMMUNITIES

- June 2021 – Spring 2025 **InitiaSciences: Founder and President**, Founding and executive direction (team of 20) of InitiaSciences, an independent non-profit organization that aims at making scientific research accessible to high school students by offering them the opportunity to participate in ongoing scientific research projects.
- 2019 – 2024 **Projet SEUR/Cap Campus Volunteer - 1000 Sciences**, In-person presentations about astrophysics in Cegeps and secondary schools in the Montreal area to encourage minority engagement with STEM.
- Nov 2021, Mar 2022 **Panelist, career event for cegep students (Online)**, Presentation about my experience, career and life path, and envisioned future pathways from a training in Physics.
- Oct 2020 **Conception of infographics: women role models in natural sciences**, Social media content describing gender inequality and impact of exposure to female role models (shared on Projet SEUR/1000 Sciences social media).
- Fall '19 – Winter '20 **Expo-Sciences project mentor**, Mentoring secondary school students for a scientific project with research and experimenting activities
- Mar 2019 **"Women in Science" / Les Filles et les Sciences: Un Duo électrisant! Kiosk**, Event encouraging young girls' interest in sciences (SEUR Project Kiosk).
- Fall 2015 **Volunteer for the Physics Student Association at UdeM**, Volunteering at the student café.
- 2026 **Invited Speaker: Empower Her Academy Workshop**, Topic: "Stellar Leadership: The Evolution and Future of Women in Space Exploration"
- 2026 **Volunteer Judge for the Southside Astronomy Art Contest**, Contest among local Chicago Elementary, Middle School and High School students. Theme: "We are made of star-stuf"
- June 2023 **Organizing and coordination of the first InitiaSciences symposium**, Entire-day event featuring the research by InitiaSciences mentees and open to the general public for pop-talks and scientific kiosks (about 200 attendees).
- 2022-2024 **Volunteer lecturer for the "Innovators" program (Technoscience - région métropolitaine nonprofit)**, Conferences about astrophysics research in secondary schools.

SCIENTIFIC COMMUNICATION AND OUTREACH ACTIVITIES

40+ PUBLIC TALKS, CLASSROOM WORKSHOPS, FESTIVALS, KIOSKS & MEDIA PRODUCTIONS; AUDIENCES FROM PRIMARY-SCHOOL PUPILS TO INTERNATIONAL HIGH-IQ SOCIETIES

Invited public talks & lectures

- 2026 **Invited Speaker: Meeting of the Chicago/Illinois American Association of Physics Teachers**, "The Exoplanet Zoo: from cotton candy planets to water worlds and the search for life".
- 2025 **Invited In-Person Presentation, American Mensa Annual Gathering**, "Unraveling Distant Worlds: From Cotton-Candy Planets to Ocean Worlds and the Search for Life". American Mensa is the U.S. chapter of the international high-IQ society Mensa.

School & classroom presentations and workshops

- April 2023 **Workshop and conference about exoplanets for the “24 hours of science” event**, Near-24h livestream featuring scientists in Quebec. During my session, dozens of classrooms were connected on the livestream channel.
- Mar 2022 **Mentor/Animator: Initiation to astrophysics research**, Series of three activities for primary school students offered to gifted kids (Gifted students program, Centre de services scolaires Marguerite-Bourgeoys - CSSMB): initiation to programming and astrophysical data
- Feb 2021 **Workshop animator: SEUR/Cap Campus - Giftedness/PGLO**, Presentation to a secondary school class about my career and life path.
- Nov 2019 **Exoplanets in the Classroom Speaker: Royal Astronomical Society of Canada pilot project**, Presentation about exoplanets for the members of the Toronto Jarvis College astronomy club (high school).
- May 2019 **Speaker: an astronomer in your classroom**, Interactive presentation in a primary school.
- May 2019 **Speaker: “Searching for Earth 2.0” – Académie de Roberval**, - Presentation about exoplanet detection and characterization methods.

Festivals, kiosks & family science events

- Nov 2023 **Volunteer at the Researchers’ Night event**, Family event; iREx kiosk, centered around transmission spectroscopy
- Sept 2023 **Volunteer at Harvest celebration**, Family event; iREx kiosk, explaining the concept of partial and total eclipse
- Nov 2022 **Researchers’ Night - Espace pour la vie, Montréal**, Scientist interacting with the general public.
- Aug 2022 **Perseid Night with the nonprofit “Chouette à voir”**, Volunteer animation of a presentation about shooting stars during the Perseid Meteor Shower.
- May 2021 **Speaker: Eurêka! Festival (Online)**, Recording of a video capsule for the “we are astronomers” project.
- Aug 2018 **Volunteer at the Astronomy Day at Campus MIL**, Organization of the event, animation of the James Webb Space Telescope kiosk.
- June 2018 **Volunteer and animator, Eurêka! Festival**, Explain astronomy context via interactive activities (family event, kiosk of the Centre de Recherche en Astrophysique du Québec).

Video, media & science-writing productions

- Sept 2023 **JWST First Year Science Conference Videoclip series**, Small outreach video about the exciting science from JWST’s first year
- 2020-2022 **Communications Intern - ExoBite videos**, Help with conception and diffusion of six videos of science popularization about exoplanets. Speaker in video 3: Extraterrestrial life.
- Oct 2021 **Recording of a video capsule about physics at UdeM**, Testimony on my experience at UdeM, for cegep students.
- June 2021 **Composi-thon and Poster - ComSciCon 2021 (Online)**, Science popularization text about my first PhD research project, and online poster on the exoplanet “Sprint de Science” I co-created with Thomas Vandal (see below).

Other engagements: judging, honors & contests

- Mar 2023 **Three-Minute Thesis contest participant (ACFAS)**,
- May 2020 **Jury member, Salon Vivre la Science et les Technologies**, Evaluation of student science projects at Saint-Laurent cegep.
- Aug 2019 **Valedictory at the graduation event**, Faculty of Arts and Sciences at UdeM.

SELECTED INTERVIEWS AND MEDIA CITATIONS

9 PRESS RELEASES, FEATURED IN THE NEW YORK TIMES, SCIENTIFIC AMERICAN, NBC NEWS, SKY & TELESCOPE, RADIO-CANADA; >30 INTERVIEWS FOR RADIO, TV, WRITTEN PRESS

- July 2026 **Interview for Sky & Telescope**, “Cotton-Candy Exoplanets: Two for the Price of One!”
- July 2026 **Press release: Piaulet-Ghorayeb et al. (2026a) results**, “Distant exoplanets may be hiding water beyond Webb Telescope’s reach”
- August 2025 **Interviews for Scientific American, and NBC News**, Commentary on TRAPPIST-1 e atmospheric reconnaissance JWST results.
Press release: Piaulet-Ghorayeb et al. (2025a) results, Results of my JWST atmospheric reconnaissance study on the temperate rocky Earth-size planet TRAPPIST-1 d. Press release relayed by NASA/STScI, ESA, CNRS, and featured in the New York Times, Universe Magazine, Universe Today, Sci.News.
- May 2025 **Feature in AAS Nova research highlights**, Results of the study led by Eva-Maria Ahrer where I led the atmosphere modeling for the interpretation of tentative atmosphere signatures on the small sub-Neptune planet GJ 3090 b.
Press release: Luque, Piaulet-Ghorayeb et al. (2025) results, Results of the study led by Rafa Luque and featuring my modeling on a revision of the evidence for biosignatures on the habitable-zone planet K2-18 b. Press release relayed by the New York Times.
- May 2025 **Radio interview for the radio show Les Années-Lumière de Radio Canada**, Communication on recent findings regarding the habitable-zone planet K2-18 b.
Press release: Piaulet-Ghorayeb et al. (2024) results, Results of my study revealing the first evidence for a planet with a water-rich “steam world” atmosphere composition. Relayed by Space.com, Science News, Discover Magazine.
- October 2024 **Curium Magazine: Interview for the Oct. 2024 edition**, Special deep-dive focused on exoplanet research in popular Québec children’s magazine.
- Apr 2023 **Special collaboration with Le Devoir journal**, Portrait of my research and outreach work.
- Dec 2022 **Interviews and citations in written press for Science & Avenir, CBC/Radio Canada, Universe Today, La Presse, le Journal de Montréal, The New York Post**, Communication on Piaulet et al. (2023) results
- Feb 2023 **Radio interviews for the Radio-Canada francophone radios tour (Moncton, Sudbury, Toronto, Vancouver, Regina), the Les Années-Lumière de Radio Canada show, and TV interviews for the Salut Bonjour show and for SETI Live**, Communication on Piaulet et al. (2023) results
- Dec 2022 **Press release: Piaulet et al. (2023) results**, Results of my study of the two candidate “water world” planets Kepler-138 c and d. PR relayed by *Hubblesite*.
- Nov 2022 **Press release: First results of the James Webb Space Telescope**, Results of the ERS team’s first paper series. PR relayed by NASA.
- Sept. 2021 **Press release: results from Pelletier et al. 2021 (incl. Piaulet)**, High-resolution spectroscopy of a hot Jupiter.
- Apr. 2021 **Interview for the Quartier Libre journal**, Exoplanet research.
- Jan-Sept 2021 **Interviews for PopSci, CBC, Supercluster, Beyond Space, Sciences et Avenir, Science et vie, the Les Années Lumière (brèves) de Radio-Canada show, The McGill Tribune, the Le Devoir journal**, Communication on Piaulet et al. (2021) results
- Jan 2021 **Press release: Piaulet et al. (2021) results**, Proposing a new formation mechanism for gas giant planets.
- Sept 2020 **Interview for the radio show Les Années Lumière de Radio-Canada**, Discussing the phosphine discovery claim in Venus’ atmosphere.
- Sept 2019 **Press release: Benneke, Wong, Piaulet, et al. (2019)**, Detection of water in the atmosphere of a habitable-zone mini-Neptune planet.

Caroline Piaulet-Ghorayeb

51 PEGASI B FELLOW (2026) & MARGARET BURBIDGE PRIZE POSTDOCTORAL FELLOW · UNIVERSITY OF CHICAGO

✉ carolinepiaulet@uchicago.edu | 🏠 cpiaulet.github.io | 📄 github.com/cpiaulet

Publications

8 1ST-AUTHOR, 63 REF'D (+9 SUBM.), 3,700+ CITATIONS, H-INDEX=29, ORCID: 0000-0002-2875-917X

[Legend: * means not peer-reviewed or ongoing review process]

- [***73**] **Piaulet-Ghorayeb**, “Sub-Neptune Atmosphere Characterization in the JWST Era” (2026, Invited Review for Nature Astronomy)
- [***72**] **Piaulet-Ghorayeb**, “STCTM: a forward modeling and retrieval framework for stellar contamination and stellar spectra” (2026, in review at JOSS, arXiv:2508.19297)
- [***71**] Adamski, H., **et al. (incl. Piaulet-Ghorayeb)**, “The Featureless JWST NIRISS/SOSS Spectrum of the Hot Super-Earth TOI-1685b Suggests Possible Sequestration of Light Elements in Core” (2026, submitted to AAS Journals)
- [***70**] Staudt, P., Ahrer, A., **Piaulet-Ghorayeb, C.** et al., “LRG-BEASTS: High-altitude aerosols in the atmosphere of WASP-69 b using WHT/ACAM and NTT/EFOSC2” (2026, in review at A&A)
- [***69**] Zhang, M., **et al. (incl. Piaulet-Ghorayeb)**, “C, N, O, S, and photochemistry in a temperate giant planet orbiting a late M dwarf” (2026, submitted to AAS Journals, arXiv:2606.23820)
- [***68**] Yang, J., **et al. (incl. Piaulet-Ghorayeb)**, “Photochemical Production of CS₂ in Temperate-to-Warm Gas Giant Exoplanet Atmospheres” (2026, submitted to AAS Journals, arXiv:2606.23820)
- [***67**] Feinstein, A., **et al. (incl. Piaulet-Ghorayeb)**, “On Linking Planet Formation Models, Protoplanetary Disk Properties, and Mature Gas Giant Exoplanet Atmospheres” (2025, submitted to AAS Journals, arXiv:2506.00669)
- [***66**] Benneke, B., Roy, P.-A., **Piaulet, C.**, et al., “JWST NIRSpec Reconnaissance Transmission Spectroscopy of the Habitable-zone Exo-Earth TRAPPIST-1g” (2024, submitted to **Nature**)
- [**65**] Kubyshkina, Egger & **Piaulet-Ghorayeb**, “Escape of Water- and Metal-enriched Atmospheres from compact Hot mini-Neptunes with CHAIN” (2026, accepted in A&A, arXiv:2606.02083)
- [**64**] **Piaulet-Ghorayeb, C.**, et al., “A Window for Water-hydrogen Demixing on Warm Metal-rich Sub-Neptunes” (2026, ApJ, 1006, 59)
- [**63**] Pelletier, S., **et al. (incl. Piaulet-Ghorayeb)**, “The breaking of cold traps and onset of titanium in ultra-hot Jupiter atmospheres: WASP-189b in context” (2026, A&A, 711, A119)
- [**62**] Frazier, R., **et al. (incl. Piaulet-Ghorayeb)**, “The Days Drag On on WASP-121 b: Interpreting Its NIRISS Spectroscopic Phase Curve with General Circulation Models” (2026, ApJ, 1004, 102)
- [**61**] Weisserman, D., **et al. (incl. Piaulet-Ghorayeb)**, “Super-Earth masses and stellar abundances from NIRPS reveal tentative evidence for water-rich formation around M dwarfs” (2026, A&A, 709, A1657)
- [**60**] Frensch, Y., **et al. (incl. Piaulet-Ghorayeb)**, “TOI-3288 b and TOI-4666 b: two gas giants transiting low-mass stars characterised by NIRPS” (2026, A&A, 707, A73)
- [**59**] Roy, P.-A., et al. (**Piaulet-Ghorayeb 5th author**), “Diversity in the haziness and chemistry of temperate sub-Neptunes” (2026, **Nature Astronomy**, 10, 371)
- [**58**] Krishnamurthy, V., Carteret, Y., **Piaulet-Ghorayeb**, et al., “Continuous helium absorption from the leading and trailing tails of WASP-107b” (2026, **Nature Astronomy**, 10, 258)
- [**57**] Lamontagne, P., **et al. (incl. Piaulet-Ghorayeb)**, “NIRPS tightens the mass estimate of GJ 3090 b and detects a planet near the stellar rotation period” (2026, A&A, 706, A278)
- [**56**] Pelletier, S., **et al. (incl. Piaulet-Ghorayeb)**, “Enriched volatiles and refractories but deficient titanium on the dayside atmosphere of WASP-121b revealed by JWST/NIRISS” (2026, A&A, 706, A2)
- [**55**] Splinter, J., **et al. (incl. Piaulet-Ghorayeb)**, “Precise Constraints on the Energy Budget of WASP-121 b from Its JWST NIRISS/SOSS Phase Curve” (2025, AJ, 170, 6, 323)
- [**54**] Allart, R., **et al. (incl. Piaulet-Ghorayeb)**, “A complex structure of escaping helium spanning more than half the orbit of the ultra-hot Jupiter WASP-121 b” (2025, **Nature Communications**, 16, 10822)

- [53] Xue, Q., **et al. (incl. Piaulet-Ghorayeb)**, “The JWST Rocky Worlds DDT Program reveals GJ 3929b to likely be a bare rock” (2025, ApJL, 995, 2, L52)
- [52] Brady, M., **et al. (incl. Piaulet-Ghorayeb)**, “An Earthlike Density for the Temperate Earth-sized Planet GJ 12 b” (2025, AJ, 170, 6, 335)
- [51] Crossfield, I., **et al. (incl. Piaulet-Ghorayeb)**, “Mapping the SO₂ shoreline in gas giant exoplanets” (2025, ApJ, 994, 184)
- [50] Parc, L., **et al. (incl. Piaulet-Ghorayeb)**, “NIRPS and TESS reveal a peculiar system around the M dwarf TOI-756: A transiting sub-Neptune and a cold eccentric giant” (2025, 2025, A&A, 702, A138)
- [49] Coulombe, L.-P., Benneke, B., Krissansen-Totton, J., L’Heureux, A., **Piaulet-Ghorayeb, C.**, et al., “Tentative evidence for the presence of volatiles on the warm super-Earth TOI-270 b” (2025, AJ, 170, 4, 226)
- [48] Bazinet, L., **et al. (incl. Piaulet-Ghorayeb)**, “Quantifying thermal water dissociation in the dayside photosphere of WASP-121 b using NIRPS” (2025, A&A, 701, A276)
- [47] Cadieux, C., L’Heureux, A., **Piaulet-Ghorayeb**, et al., “Detailed Architecture of the L 98-59 System and Confirmation of a Fifth Planet in the Habitable Zone” (2025, AJ, 170, 154)
- [46] **Piaulet-Ghorayeb**, et al., “Strict Limits on Potential Secondary Atmospheres on the Temperate Rocky Exo-Earth TRAPPIST-1 d” (2025, ApJ, 989, 181)
- [45] Luque, R., **Piaulet-Ghorayeb**, et al., “Insufficient evidence for DMS and DMDS in the atmosphere of K2-18 b: From a joint analysis of JWST NIRISS, NIRSpec, and MIRI observations” (2025, A&A, 700, A284)
- [44] Gomes da Silva, J., **et al. (incl. Piaulet-Ghorayeb)**, “Blind search for activity-sensitive lines in the near-infrared using HARPS and NIRPS observations of Proxima and Gl 581” (2025, A&A, 700, A177)
- [43] Suarez Mascareño, A., **et al. (incl. Piaulet-Ghorayeb)**, “Diving into the planetary system of Proxima with NIRPS: Breaking the metre per second barrier in the infrared” (2025, A&A, 700, A11)
- [42] Bouchy, F., **et al. (incl. Piaulet-Ghorayeb)**, “NIRPS joining HARPS at ESO 3.6 m: On-sky performance and science objectives” (2025, A&A, 700, A10)
- [41] Vulato, V., **et al. (incl. Piaulet-Ghorayeb)**, “Hydride ion continuum hides absorption signatures in the NIRPS near-infrared transmission spectrum of the ultra-hot gas giant WASP-189b” (2025, A&A, 700, A9)
- [40] Allart, R., **et al. (incl. Piaulet-Ghorayeb)**, “NIRPS detection of delayed atmospheric escape from the warm and misaligned Saturn-mass exoplanet WASP-69 b” (2025, A&A, 700, A7)
- [39] Taylor, J., **et al. (incl. Piaulet-Ghorayeb)**, “JWST NIRISS transmission spectroscopy of the super-Earth GJ 357b, a favourable target for atmospheric retention” (2025, MNRAS, 540, pp. 3677-3692)
- [38] Fournier-Tondreau, M., **et al. (incl. Piaulet-Ghorayeb)**, “Transmission spectroscopy of WASP-52 b with JWST NIRISS: water and helium atmospheric absorption, alongside prominent star-spot crossings” (2025, MNRAS, 539, pp. 422-438)
- [37] Ahrer, E.-M., Radica, M., **Piaulet-Ghorayeb**, et al., “Escaping Helium and a Highly Muted Spectrum Suggest a Metal-enriched Atmosphere on Sub-Neptune GJ 3090 b from JWST Transit Spectroscopy” (2025, ApJL, 985, L10)
- [36] Morel, K., **et al. (incl. Piaulet-Ghorayeb)**, “A Moderate Albedo from Reflecting Aerosols on the Dayside of WASP-80 b Revealed by JWST/NIRISS Eclipse Spectroscopy” (2025, AJ, 169, 277)
- [35] Monaghan, C., **et al. (incl. Piaulet-Ghorayeb)**, “Low 4.5 μ m Dayside Emission Disfavors a Dark Bare-rock Scenario for the Hot Super-Earth TOI-431 b” (2025, AJ, 169, 239)
- [34] Radica, M., **Piaulet-Ghorayeb, C.**, et al., “Promise and Peril: Stellar Contamination and Strict Limits on the Atmosphere Composition of TRAPPIST-1c from JWST NIRISS Transmission Spectra” (2025, ApJL, 979, L5)
- [33] Agol, E., **et al. (incl. Piaulet-Ghorayeb)**, “Updated Forecast for TRAPPIST-1 Times of Transit for All Seven Exoplanets Incorporating JWST Data” (2024, RNAAS, 8, 879-898, Vol. 8, 10, 274)
- [32] **Piaulet-Ghorayeb, C.**, et al., “JWST/NIRISS reveals the water-rich “steam world” atmosphere of GJ 9827 d” (2024, ApJL, 974, L10)
- [31] Bell, T., **et al. (Piaulet: Tier 3)**, “Nightside clouds and disequilibrium chemistry on the hot Jupiter WASP-43b” (2023, **Nature Astro.**, 8, 879-898, Volume 8, p. 879-898)

- **[30] JWST TRAPPIST-1 Community Initiative (incl. Piaulet)**, “A roadmap to the efficient and robust characterization of temperate terrestrial planet atmospheres with JWST” (2024, **Nature Astro.**, 8, 810-818)
- **[29] Cadieux, C. et al. (incl. Piaulet-Ghorayeb)**, “Transmission Spectroscopy of the Habitable Zone Exoplanet LHS 1140 b with JWST/NIRISS” (2024, ApJL, 970, L2)
- **[*28] Artigau, E. and NIRPS Consortium (incl. Piaulet)**, “NIRPS first light and early science: breaking the 1 m/s RV precision barrier at infrared wavelengths” (2024, SPIE Proceedings, arXiv:2406.08304)
- **[*27] Benneke, B., Roy, P.-A., Coulombe, L.-P., Radica, M., Piaulet, C., et al.**, “JWST Reveals CH₄, CO₂, and H₂O in a Metal-rich Miscible Atmosphere on a Two-Earth-Radius Exoplanet” (2024, submitted to ApJL, arXiv: 2403.03325)
- **[26] Fournier-Tondreau, M., MacDonald, R.J., Radica, M., Lafrenière, D., Welbanks, L., Piaulet, C., et al.**, “Near-Infrared Transmission Spectroscopy of HAT-P-18 b with NIRISS: Disentangling Planetary and Stellar Features in the Era of JWST” (2024, MNRAS, 528, 3354)
- **[25] Howard, W. S., et al. (incl. Piaulet)**, “Characterizing the Near-Infrared Spectra of flares from TRAPPIST-1 during JWST Transit Spectroscopy Observations” (2023, ApJ, 959, 64)
- **[24] Lim, O., Benneke, B., Doyon, R., Macdonald, R. J., Piaulet, C., et al.**, “Atmospheric Reconnaissance of TRAPPIST-1 b with JWST/NIRISS: Evidence for Strong Stellar Contamination in the Transmission Spectra” (2023, ApJL, 955, L22)
- **[23] Roy, P.-A., Benneke, B., Piaulet, C., et al.**, “Water absorption in the transmission spectrum of the water-world candidate GJ 9827 d” (2023, ApJL, 954, L52)
- **[22] Coulombe, L.-P., et al. (Piaulet: Tier 3)**, “A broadband thermal emission spectrum of the ultra-hot Jupiter WASP-18b” (2023, **Nature**, 620, 292)
- **[21] Albert, L., et al. (incl. Piaulet)**, “The Near Infrared Imager and Slitless Spectrograph for the James Webb Space Telescope – III. Single Object Slitless Spectroscopy” (2023, PASP, 135, 075001)
- **[20] Peterson, M., Benneke, B., Collins, K., Piaulet, C., et al.**, “A temperate Earth-sized planet with tidal heating transiting an M6 star” (2023, **Nature**, 617, 701) **Note: Benneke & Piaulet lead authors after M. Peterson left the field in 2021.**
- **[19] Feinstein, A., et al. (Piaulet: Tier 3)**, “Early Release Science of the exoplanet WASP-39b with NIRISS-SOSS” (2023, **Nature** 614, 670–675)
- **[18] Ahrer, E.-M., et al. (Piaulet: Tier 3)**, “Early Release Science of the exoplanet WASP-39b with JWST NIRCам” (2023, **Nature** 614, 653–658)
- **[17] Alderson, L., et al. (Piaulet: Tier 3)**, “Early Release Science of the exoplanet WASP-39b with JWST NIRSpec G395H” (2023, **Nature** 614, 664–669)
- **[16] Rustamkulov, Z., et al. (Piaulet: Top Tier 1/4)**, “Early Release Science of the exoplanet WASP-39b with JWST NIRSpec PRISM” (2023, **Nature** 614, 659–663)
- **[15] The JWST Transiting Exoplanet Community Early Release Science Team**, “Identification of carbon dioxide in an exoplanet atmosphere” (2023, **Nature** 614, 649–652)
- **[14] Piaulet, C., et al.** “Evidence for the volatile-rich composition of a 1.5-Earth-radius planet” (2023, **Nature Astro.** 7, 206–222)
- **[13] Roy, P.-A., Benneke, B., Piaulet, C., et al.**, “Is the Hot, Dense Sub-Neptune TOI-824 b an Exposed Neptune Mantle? Spitzer Detection of the Hot Dayside and Reanalysis of the Interior Composition” (2022, ApJ, 941, 89)
- **[12] Bell, T., et al. (incl. Piaulet: Tier 2)**, “Eureka!: An End-to-End Pipeline for JWST Time-Series Observations” (2022, JOSS, 7, 4503)
- **[11] Radica, M., et al. (incl. Piaulet)**, “APPLESOSS: A Producer of Profiles for SOSS. Application to the NIRISS SOSS Mode” (2022, PASP, 134, 104502)
- **[10] Darveau-Bernier, A., et al. (incl. Piaulet)**, “ATOCA: an Algorithm to Treat Order Contamination. Application to the NIRISS SOSS Mode” (2022, PASP, 134, 094502)
- **[9] Morris, B. M., Heng, K., Jones, K., Piaulet, C., et al.**, “Physically-motivated basis functions for temperature maps of exoplanets” (2022, A&A, 660, A123)
- **[8] Zeng, L., Jacobsen, S. B., Hyung, E., Levi, A., Nava, C., Kirk, J., Piaulet, C., et al.**, “New Perspectives on the Exoplanet Radius Gap from a Mathematica Tool and Visualized Water Equation of State” (2021, ApJ, 923, 247)

- **[7]** Pelletier, S., Benneke, B., Darveau-Bernier, A., Boucher, A., Cook, N. J., **Piaulet, C.**, et al., “Where Is the Water? Jupiter-like C/H Ratio but Strong H₂O Depletion Found on τ Boötis b Using SPIRou” (2021, AJ, 162, 73)
- **[6]** **Piaulet, C.**, et al., “WASP-107b’s density is even lower: a case study for the physics of planetary gas envelope accretion and orbital migration” (2021, AJ, 161, 70)
- **[5]** Kosiarek, M., Berardo, David A., Crossfield, I. J. M., Laguna, C., **Piaulet, C.**, et al., “Physical Parameters of the Multi-Planet Systems HD 106315 and GJ 9827” (2021, AJ, 161, 47)
- **[4]** St-Louis, L., **Piaulet, C.**, et al., “An extensive spectroscopic time series of three Wolf-Rayet stars - II. A search for wind asymmetries in the dust-forming WC7 binary WR137” (2020, MNRAS, 497, 4448)
- **[3]** Benneke, B., Wong, I., **Piaulet, C.**, et al., “Water Vapor and Clouds on the Habitable-zone Sub-Neptune Exoplanet K2-18b” (2019, ApJL, 887, L14)
- **[2]** Ramaramanantsoa, T. **et al. (incl. Piaulet)**, “BRITE-Constellation high-precision time-dependent photometry of the early-O-type supergiant ζ Puppis unveils the photospheric drivers of its small- and large-scale wind structures” (2018, MNRAS, 473, 5532)
- **[*1]** **Piaulet, C.**, Mochizuki, K., Arguin, J.-F., “Improvement of the multivariate analysis method for the reduction of electron charge misidentification in ATLAS data” (2017, ATL-COM-PHYS-2017-1539)